

Sri Lanka Institute of Information Technology



Application Frameworks – SE3040

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Assignment – Testing Report

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1. Objective

This report documents the end-to-end testing progress and results for SanTrack backend and frontend modules, including functional workflows, API correctness, validation behavior, security controls, third-party integrations, and UI behavior.

2. Scope

The following areas are covered:

- Functional Testing
- API Testing (Postman)
- Validation Testing
- Authentication and Authorization Testing
- Third-Party API Testing
- UI Testing

3. Test Environment

- OS : Windows
- Backend : Node.js + Express
- Frontend : React (Vite)
- Database : MongoDB
- API Base URL: `http://localhost:5000/api`
- Tools : Postman, Browser DevTools, VS Code terminal

4. Test Accounts and Roles

- Admin account : Full management access
- Inspector account : Inspection and issue reporting access
- Community member account : Issue reporting and limited read access

5. Functional Testing

5.1 Test Cases

Test ID	Scenario	Preconditions	Steps	Expected Result	Actual Result	Status	Evidence
FT-01	Register user	Backend running	Submit valid registration data	User created , 201 response	User created and returned with profile fields	Pass	Screenshot/JSON
FT-02	Login user	Registered user exists	Submit valid email/password	Token returned, 200 response	Token and user object returned	Pass	Screenshot/JSON
FT-03	Create village	Admin token available	POST village with valid payload	Village created , 201	Village created successfully	Pass	Screenshot/JSON
FT-04	Create inspection	Inspector token and facilityId available	POST inspection with valid score	Inspection created , 201	Inspection saved with computed status	Pass	Screenshot/JSON
FT-05	Submit issue	Inspector or community token, facilityId	POST issue with valid payload	Issue created with pending status	Issue created and stored	Pass	Screenshot/JSON
FT-06	Update issue status	Admin token, issueId exists	PUT issue status to in_progress/resolved	Status updated, 200	Status changed and persisted	Pass	Screenshot/JSON
FT-07	Delete record	Admin token, target record exists	DELETE village/inspection/issue by id	Record deleted, 200	Record removed with success message	Pass	Screenshot/JSON

5.2 Functional Testing Notes

- Core workflow from registration to issue lifecycle is functional.
- Status transition for issues is handled by admin role.
- Deletion endpoints return success messages and remove records as expected.

6. API Testing (Postman)

6.1 API CRUD Coverage

Resource	Create (POST)	Read All (GET)	Read By ID (GET)	Update (PUT)	Delete (DELETE)	Result
Auth	Register/Login/Google	Profile	N/A	Profile update	Account delete	Pass
Users	Yes	Yes	Yes	Yes	Yes	Pass
Villages	Yes	Yes	Yes	Yes	Yes	Pass
Facilities	Yes	Yes	Yes	Yes	Yes	Pass
Inspections	Yes	Yes	Yes	Yes	Yes	Pass
Issues	Yes	Yes	Yes	Yes	Yes	Pass

6.2 Protected Route Testing

Test ID	Endpoint	Token	Expected	Actual	Status
API-PR-01	GET /villages	Missing	401 Unauthorized	401 returned	Pass
API-PR-02	POST /villages	Inspector token	403 Forbidden	403 returned	Pass
API-PR-03	POST /villages	Admin token	201 Created	201 returned	Pass
API-PR-04	PUT /issues/:id	Community token	403 Forbidden	403 returned	Pass
API-PR-05	PUT /issues/:id	Admin token	200 Success	200 returned	Pass

6.3 Invalid Input Testing

Test ID	Endpoint	Invalid Input	Expected	Actual	Status
API-IV-01	POST /auth/register	invalid email	400 validation error	400 returned	Pass
API-IV-02	POST /villages	gps.lat = 200	400 validation error	400 returned	Pass

API-IV-03	POST /inspections	score = 20	400 validation error	400 returned	Pass
API-IV-04	PUT /issues/:id	status = closed	400 validation error	400 returned	Pass

6.4 Role Restriction Testing

Test ID	Endpoint	Role	Expected	Actual	Status
API-RR-01	GET /users	inspector	403	403 returned	Pass
API-RR-02	GET /users	admin	200	200 returned	Pass
API-RR-03	POST /inspections	community	403	403 returned	Pass
API-RR-04	POST /inspections	inspector	201	201 returned	Pass

7. Validation Testing

Test ID	Module	Validation Case	Input	Expected	Actual	Status
VT-01	Auth	Missing required fields	name/password omitted	400 + field messages	400 returned with error list	Pass
VT-02	Auth	Invalid email	email=abc	400	400 returned	Pass
VT-03	Village	Invalid GPS lat	lat=100	400	400 returned	Pass
VT-04	Village	Invalid GPS lng	lng=300	400	400 returned	Pass
VT-05	Inspection	Invalid score	score=0 or 11	400	400 returned	Pass
VT-06	Issue	Invalid status enum	status=closed	400	400 returned	Pass
VT-07	File type	Invalid file type upload	non-image file	Rejected	Not applicable in current API upload flow	N/A

Validation Note on File Type

Current backend validates inspection images as string URLs and does not expose a dedicated multipart upload endpoint in current route set. File-type rejection test is therefore marked “N/A” for current implementation.

8. Authentication and Authorization Testing

Test ID	Scenario	Endpoint	Expected	Actual	Status
AA-01	Route without token	GET /inspections	401	401 returned	Pass
AA-02	Wrong role	DELETE /inspections/:id with inspector token	403	403 returned	Pass
AA-03	Valid role access	DELETE /inspections/:id with admin token	200	200 returned	Pass
AA-04	Invalid token	GET /profile with malformed token	401	401 returned	Pass
AA-05	Valid token	GET /profile with valid token	200	200 returned	Pass

9. Third-Party API Testing

Test ID	Integration	Test	Expected	Actual	Status	Notes
TP-01	Google OAuth	POST /auth/google with valid credential	User logged in/created	Login success with token	Pass	Uses Google token verification service
TP-02	Reverse Geocode	GET /villages/reverse-geocode with valid lat/lng	Address payload returned	Address fields returned	Pass	Admin role required
TP-03	Email Notification	Create issue then update status	Email sent to reporter	Triggered where SMTP config available	Partial	Depends on EMAIL_USER/EMAIL_PASS

TP-04	Cloudinary Upload	Upload image and store URL	Upload succeeds and URL saved	No dedicated upload endpoint verified	N/A	Marked as implementation gap
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10. UI Testing

Test ID	Area	Test	Expected	Actual	Status
UI-01	Navigation	Public and dashboard routes navigation	Correct page transitions	Navigation works	Pass
UI-02	Forms	Login, signup, issue, inspection forms submit	Data sent and success feedback shown	Forms submit and handle responses	Pass
UI-03	Dashboard	Admin/Inspector/Community dashboards load	Correct widgets and access	Dashboards load by role	Pass
UI-04	Responsive	Mobile and desktop layout check	Usable layout without major breakage	Basic responsive behavior verified	Pass

11. Overall Result Summary

- Total test cases planned : 31
- Passed : 28
- Failed : 0
- Partial : 1
- N/A : 2

Conclusion

SanTrack core modules are stable for functional, API, validation, auth/authorization, and UI baseline checks. Role restrictions and protected routes behave correctly. Third-party integrations for Google OAuth and reverse geocoding are functional. Email notifications are conditionally functional based on environment configuration. Cloudinary upload flow requires explicit endpoint-level implementation/testing to fully satisfy file upload validation requirements.